

2. Hepatic caecae

The hepatic caecae are also termed 'midgut caecae'. They contain secretory and absorptive cells.

3. Glandular cells of the mesenteron

The glandular cells of the mesenteron secrete enzymes such as **maltase**, **invertase**, **proteases** and **lipase**.

III. Physiology of Digestion

- a) **Food collection:** Cockroach is an **omnivorous** insect. It feeds on all types of organic matter. It locates the food by the olfactory sensillae of antenna, labial palps and maxillary palps. The food is seized with the help of forelegs, labrum and labium. It is passed to the mandibles for biting and chewing. The laciniae, galeae, glossae and paraglossae hold the food during chewing. The labrum and labium prevent the food from falling down. The food is mixed with saliva during mastication.
- b) **Digestion:** After swallowing, the food passes through the pharynx and oesophagus, and reaches the crop. In the crop, food is mixed with digestive juices that are regurgitated into it through the grooves of the gizzard. Hence, most of the food is digested in the crop. The partly digested food is filtered by the bristles of the gizzard and later it passes through the stomodeal valve into the ventriculus.

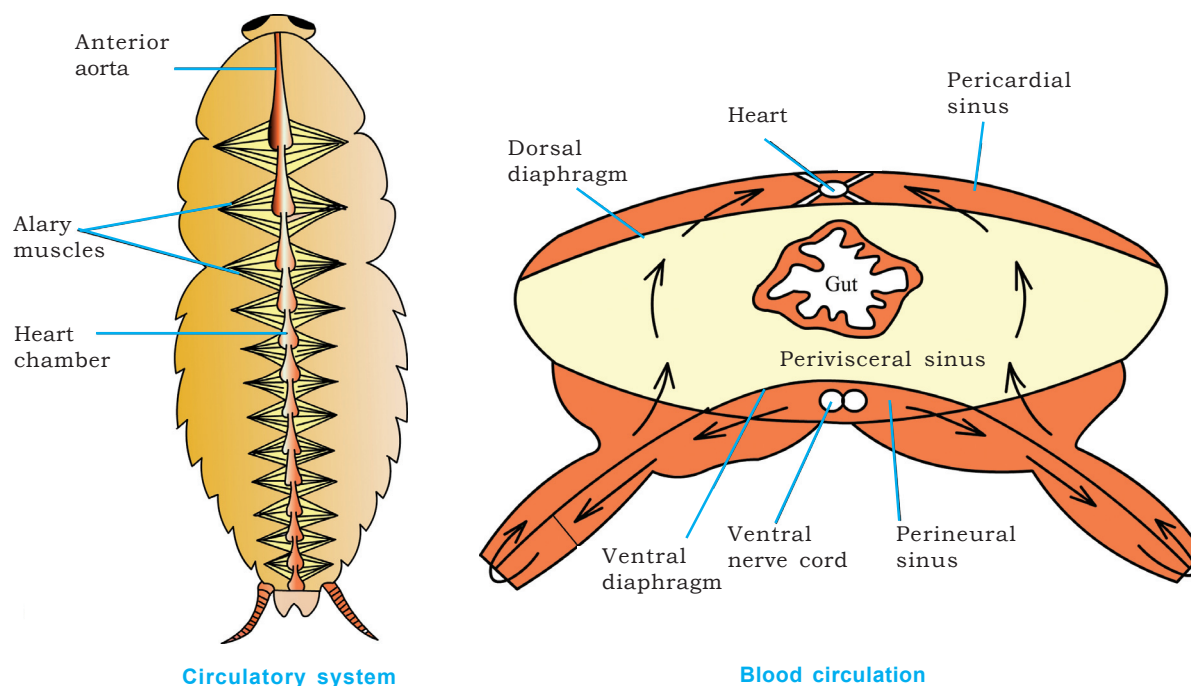
The enzyme amylase of the salivary juice converts starches into **disaccharides**. **Invertase** or sucrase digests sucrose into **glucose** and **fructose**. Maltase converts maltose into **glucose**. The enzyme lipase digests lipids into **fatty acids** and **glycerol**. Proteases digest proteins into amino acids. Cellulose of the food is digested by the enzyme **cellulase** secreted by the microorganisms present in the hindgut of cockroach. Cellulose is converted into glucose.

In the ventriculus, the digested food is absorbed. The undigested food is passed into the ileum, colon, and then reaches the rectum, where water is reabsorbed by rectal papillae. Then the remaining material is finally **defaecated** as **dry pellets**, through the anus.

The crop of *Periplaneta* does not secrete any digestive enzymes, but most of the food is digested in it. Explain how it is possible?

7.5 CIRCULATORY SYSTEM OF *PERIPLANETA*

The circulatory system helps in the transportation of digested food, hormones etc., from one part to another in the body. *Periplaneta* has an **open type** of circulatory system as the blood, or haemolymph, flows freely within the body cavity or haemocoel. Blood vessels are poorly developed and open into spaces. Visceral organs located in

Fig. 7.6 *Periplaneta* : Circulatory system

the haemocoel are bathed in the haemolymph. The three main parts associated with the blood circulatory system of *Periplaneta* are the haemocoel, heart, and blood.

1. Haemocoel: The haemocoel of cockroach is divided into three sinuses by two muscular, horizontal membranes, called **dorsal diaphragm** or 'pericardial septum' and **ventral diaphragm**. Both the diaphragms have valvular pores. There is a series of paired triangular muscles, called **alary muscles**. Every segment has one pair of these muscles situated on the lateral sides of the body. These are attached to the pericardial septum by their broad bases and to the terga by their pointed ends or apices. The three sinuses of the haemocoel are known as **pericardial haemocoel** or the '**dorsal sinus**', the **perivisceral haemocoel** or the '**middle sinus**' and **sternal haemocoel** or '**ventral sinus**' or '**perineural sinus**'. The middle sinus is very large as it encloses most of the visceral organs. The dorsal and ventral sinuses are small in size and they enclose **heart** and **nerve cord**, respectively.

2. Heart

The heart lies in the pericardial haemocoel or dorsal sinus. It is a long, muscular, contractile tube found along the mid dorsal line, beneath the terga of the thorax and abdomen. It consists of 13 chambers. Every chamber opens into the other present in front of it through valvular openings. Three of the thirteen chambers are situated in the thorax and ten in the abdomen. Its posterior end is closed while the anterior end

is continued forward as the anterior **aorta**. At the posterior side of each chamber, except the last, there is a pair of small apertures called '**ostia**' one on each side. Ostia have valves which allow the blood to pass only into the heart from the dorsal sinus.

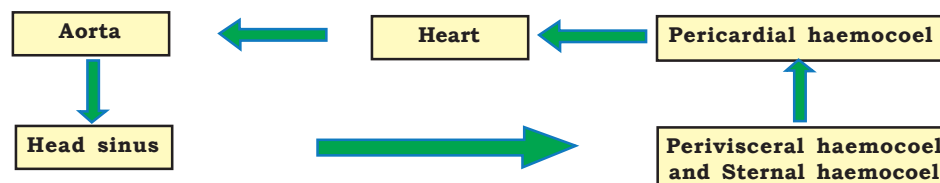
3. Circulatory fluid

The circulatory fluid of *Periplaneta* is colourless and is called **haemolymph**. It consists of a fluid called plasma, and free corpuscles or haemocytes, which are 'phagocytic'. The phagocytes are large in size and can 'ingest' foreign particles such as bacteria. There is no respiratory pigment in the blood and so it plays no major role in respiration. The important functions of the blood are:

1. It absorbs digested food from the alimentary canal and distributes it to the rest of the body.
2. It brings nitrogenous wastes from all parts of the body to the excretory organs for their elimination.
3. It carries defensive phagocytes to the places of infection where they engulf the germs and disintegrating tissue parts.
4. It transports secretions of the ductless glands to the target organs.

4. Circulation of blood

The blood flows forward in the heart by the contractions of its chambers. At the anterior end of the heart, the blood flows into the aorta and from there it enters the sinus of the head. From the head sinus, the blood flows into the perivisceral and sternal sinuses. On contraction of the alary muscles the pericardial septum is pulled down. This increases the volume of the pericardial sinus. Hence blood flows from the perivisceral sinus into the pericardial sinus through the apertures of the pericardial septum. On relaxation of the alary muscles, the pericardial septum moves upwards to its original position. This forces the blood, to enter the chambers of the heart through the ostia from the pericardial sinus.



Course of circulation of blood in *Periplaneta*

7.6 RESPIRATORY SYSTEM OF *PERIPLANETA*

Due to the absence of respiratory pigment, the blood of cockroach is colourless and it cannot carry oxygen to different tissues. Therefore a tracheal system is developed to carry the air directly to the tissues. The respiratory system of cockroach consists of stigmata, tracheae and tracheoles.